EE7500 : Assignment - 1.

1. $Z_0 = 50\,\Omega \quad f = 1\,GHz \quad Z_L = 75 + j50\,\Omega$

$l = 0.15\lambda \quad C = \lambda f \quad \lambda = \frac{3\times10^8}{10^9} = 0.3m.$

a) $\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{75 + j50 - 50}{125 + j50} = 0.31 + j0.275$

$\Gamma_L = 0.415 \angle 41.6^\circ$

$VSWR = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} \Rightarrow 2.42$

b) $\frac{Z_{in}}{Z_0} = \frac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}$

$= \frac{75 + j50 + j50\tan\left(\frac{2\pi}{\lambda} \cdot 0.15\lambda\right)}{50 + j(75+j50)\tan\left(\frac{2\pi}{\lambda} \cdot 0.15\lambda\right)}$

$\Rightarrow 0.9858 - 0.906j$

$Z_{in} = 49.29 - 45.3j$

c) $P_{inc} = 10W$

$P_{refl} = |\Gamma|^2 \times 10$

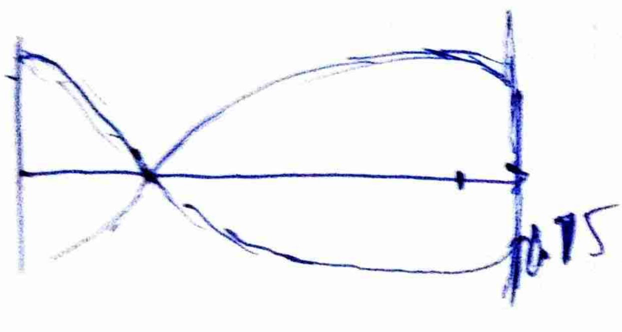
$= (0.415)^2 \times 10 = 1.722W.$

$P_{deli} = 10 - P_{refle} = 8.27W.$

d) $|V(d)| = |V_0^+|(1 + e^{2j(\beta z)}\Gamma)| \Rightarrow |V_0^+||1 + e^{j(\theta + 2\beta l)}|$

$V_{max} = |V_0^+| \, |1 + |\Gamma||$ $\quad V_{min} = |V_0^+||(1 - |\Gamma|)|.$

$= |V_0^+|(1.415)$ $\quad V_{min} = |V_0^+|(0.585)$



$d_{max} \Rightarrow$

$\theta + 2\beta l = 2\pi$

$l \Rightarrow \frac{2\pi - \theta}{2\beta}$

$l \Rightarrow 0.132\ m$

$= 0.44\lambda.$

$d_{min} \Rightarrow \theta + 2\beta l = \pi$

~~$2\beta l \Rightarrow \pi - \theta$~~

$2\beta l = (\pi - \theta)$

$l \Rightarrow \frac{\pi - \theta}{2\beta} \Rightarrow 0.057 = 0.19\lambda$

2.a). $Z_L = 100\,\Omega$, $Z_0 = 50\,\Omega$. $f_0 = 2.4\,GHz$,
$\varepsilon_r = 4.4$ (FR-4).

$$Z_0' = \sqrt{Z_0 Z_L} = 70.71$$

$$Z_{QW} = 70.71\,\Omega.$$

b) $$l = \frac{\lambda'}{4} = \frac{3\times10^8}{2.4\times10^9 \times 4\sqrt{\varepsilon_r}} \Rightarrow \frac{31.25\ \text{mm}}{\sqrt{\varepsilon_r}}$$

$$\lambda' = \frac{\lambda_0}{\sqrt{\varepsilon}} \Rightarrow 14.89\ \text{mm}$$

c). $$\Gamma_{in} \Rightarrow \frac{Z_{in}-Z_0}{Z_{in}+Z_0}, \quad Z_{in} = Z_0'\left(\frac{Z_L + jZ_0'\tan(\beta l)}{Z_0' + jZ_L\tan\beta l}\right)$$

$$Z_{in} = 70.71\left(\frac{100 + j70.71\tan\beta l}{70.71 + j100\tan\beta l}\right)$$

$$|\Gamma_{in}| < 0.1$$

$$\frac{Z_{in}-Z_0}{Z_{in}+Z_0} < 0.1 \qquad 0.9 Z_{in} < 1.1 Z_0$$

$$Z_{in} < \frac{1.1 Z_0}{0.9}$$

2) c) $70.71\left(\dfrac{100 + j70.71\tan\beta l}{70.71 + j100\tan\beta l}\right) < \dfrac{1.1}{0.9}(50)$.

$$\left|\frac{100 + j70.71\tan\beta l}{70.71 + j100\tan\beta l}\right|^2 < (0.864)^2 = 0.7464.$$

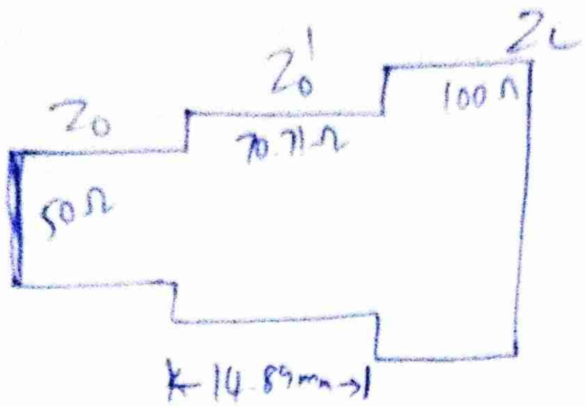
$$100^2 + 70.71^2\tan^2\beta l < 0.7464\,(70.71^2 + 100^2\tan^2\beta l)$$

$$6268.071 < 2464\tan^2\beta l$$

$$\tan\beta l > 1.59$$

$$\beta l > 57.9$$

d)



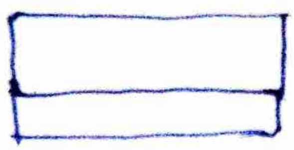
3. $f = 200\,MHz$, Layer 1 : Al, $t_1 = 100\,\mu m$

$\sigma_{Al} = 3.5 \times 10^7\ S/m$

$\mu_r = 1$

Layer 2 : Cu; $t_2 = 50\,\mu m$

$\sigma_{Cu} = 5.8 \times 10^7\ S/m$



a) $\delta_s = \dfrac{1}{\sqrt{\pi f \mu_0 \sigma}}$ at 200 MHz

$$\delta_{s_{Al}} = \frac{1}{\sqrt{\pi \times 200 \times 10^6 \times 4\pi \times 10^{-7} \times 3.5 \times 10^7}}$$

$\Rightarrow 6.01\ \mu m.$

$$\delta_{s_{Cu}} = \frac{1}{\sqrt{\pi \times 200 \times 10^6 \times 4 \pi \times 5.8}} \Rightarrow 4.67\ \mu m$$

$L_{Al} \Rightarrow \frac{100\ \mu m}{6.01\ \mu m} = 16.63$ skin depth.

$L_{Cu} = \frac{50\ \mu m}{4.67\ \mu m} = 10.69 \approx 10.7$ skin depths

$SE \approx 8.686\left(\frac{t}{\delta_s}\right)$ dB.

$SE_{AL} = 8.686 \times 16.63 = 144.47$ dB.

$\therefore SE_{Cu} = 8.686 \times 10.7 = 92.94$ dB.

$SE_{tot} = 237.41$ dB.

b) $L_{Al} = \frac{150\ \mu m}{6.01\ \mu m} = 24.95$ skin depth.

$SE_{Al} = 8.686 \times 24.95 = 216.7$ dB.

The shielding effectiveness of prior model is greater than S.E AL
and b2, Cu is stronger conductor
$\therefore$ smaller skin depth $\therefore$ with same thickness of the material greather attenuation is attained.

c). $f = 5 GHz.$ $H_0 = 1$ A/m.   Cu Slab.

$Z_s = \frac{1+j}{\sigma \delta_s}$ $\frac{\vec{E}_{tan}}{\vec{H}_{tan}} = Z_s$.
↳ surface impedance.

~~E_{s0}~~ $\vec{E}_{tan}$ ⇒ $Z_s = \frac{1+j}{5.8 \times 10^7 \times} \times \sqrt{\pi \times 5 \times 10^9 \times 4\pi \times 10^{-7} \times 5.8 \times 10^7}$

$Z_s = 0.0184 + 0.0184j$

~~Z_s~~ =

$\vec{E}_{tan} = Z_s \times H_{tan} \Rightarrow (0.0184 + j0.0183)$ V/m

d). $E(z) = E_0 e^{-z/\delta_s}$.

at ~~e~~ $E(t) = E_0 e^{-t/\delta_s}$. $e^{-t/\delta_s} << 1$).

$e^{-2} \approx 0.13$
$e^{-3} \approx 0.05$
$e^{-5} \approx 0.0067$.

$\frac{t}{\delta_s} >> 1$

$\boxed{\therefore t >> \delta_s}$

∴ yes, valid here

4). $f = 10\,GHz \quad +z$

$\vec{E} = (a_x \hat{x} + a_y e^{j\delta} \hat{y}) e^{-jk_0 z}$ V/m.

a) $a_x = 3, \; a_y = 4 \quad \delta = 0$

$\vec{E} = (3\hat{x} + 4\hat{y}) e^{-jk_0 z} e^{j\omega t} \Rightarrow$ linear polarization.

$\tan\psi \Rightarrow \pm \frac{4}{3}. \qquad \psi = \tan^{-1}(\frac{4}{3}) \Rightarrow 53.1°$

$|\vec{S}_{av}| = \frac{1}{2\eta} |E|^2 \Rightarrow \frac{1}{2(120\pi)} (25) \Rightarrow 0.033$ W/m2

b) $a_x = 3, \; a_y = 4, \; \delta = -\pi/2 \qquad \vec{E}(z=0,t).$

$t = 0, \; T/4, \; T/2, \; 3T/4$

$\vec{E}(0,t) \Rightarrow (3\hat{x} + 4 e^{-j\pi/2} \hat{y}) e^{j\omega t}$ V/m.

$\vec{E}(0,0) \Rightarrow 3\hat{x} + 4e^{-j\pi/2}\hat{y}.$

$\vec{E}(0,T/4) \Rightarrow (3\hat{x} + 4e^{-j\pi/2}\hat{y}) e^{j\frac{2\pi}{T}\cdot\frac{T}{4}} \Rightarrow$

$\Rightarrow 3e^{j\frac{\pi}{2}}\hat{x} + 4\hat{y}.$

$f = \frac{1}{T}$

$\omega = 2\pi f.$

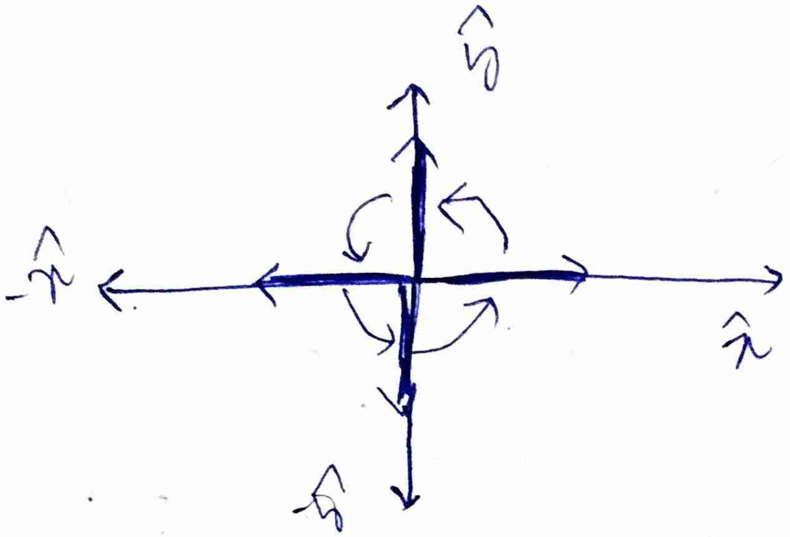
$= \frac{2\pi}{T}$

$\vec{E}(0,T/2) \Rightarrow (3\hat{x} + 4e^{-j\pi/2}\hat{y}) e^{j\frac{2\pi}{T}\cdot\frac{T}{2}} \Rightarrow -3\hat{x} - 4e^{-j\pi/2}\hat{y}.$

$$\vec{E}(0, T/4) = \left(3\hat{x} + 4e^{-j\pi/2}\hat{y}\right) e^{j\frac{2\pi}{T}\cdot\frac{3T}{4\cdot2}}$$

$$\Rightarrow 3e^{-j\pi/2}\hat{x} - 4\hat{y}.$$

$a_x \mp a_y.$



(a) $E(0,t) = 3\cos\omega t\,\hat{x} + 4\sin\omega t\,\hat{y}.$

$E(0,0) = 3\cos w(0)\,\hat{x} = 3\hat{x}.$

$E(0, T/4) = 3\cos\left(\frac{\pi}{2}\right)\hat{x} + 4\sin\left(\frac{\pi}{2}\right)\hat{y} \Rightarrow 4\hat{y}.$

$E(0, T/2) = 3\cos(\pi)\,\hat{x} + 4\sin(\pi)\,\hat{y} \Rightarrow -3\hat{x}$

$E\left(0, \frac{3T}{4}\right) \Rightarrow$ ~~[illegible]~~ $+ 3(0)\,\hat{x} - 4\hat{y} \Rightarrow -4\hat{y}$

Left eliptical polarization.

c) $\vec{E}_1 = 2\hat{x}\, e^{-jk_0 z}$ V/m

$\vec{E}_2 = 3\hat{x}\, e^{-j(k_0 z + \pi/3)}$ V/m $\rightarrow$ aditional phase $\pi/3$

$$|S_1| = \frac{1}{2\eta}|2|^2 = \frac{2}{\eta}$$

$$|S_2| = \frac{1}{2\eta}|3|^2 = \frac{9}{2\eta}$$

$$\vec{E} = e^{-jk_0 z}\left(2 + 3e^{-j\frac{\pi}{3}}\right)\hat{x}$$

$$|\vec{E}|^2 = \left(4 + 9 + 2(6)\cos\left(-\frac{\pi}{3}\right)\right)$$

$$\cancel{\Rightarrow 13 + 12} \Rightarrow 19$$

$$|S|_{tot} \Rightarrow \frac{1}{2\eta}\, 19 \Rightarrow \frac{19}{2\eta}$$

$$|S_1| + |S_2| = \frac{13}{2\eta} \qquad \therefore |S|_{tot} \neq |S_1| + |S_2|$$

Power scales with $|E|^2$ and not just E

d) $\vec{E}_1 = 2\hat{x}\, e^{-jk_0 z}$ V/m. horizontally pola.) direct

$\vec{E}_2 = 1.5\,\hat{y}\, e^{-j(k_0 z + \frac{\pi}{4})}$ V/m veritically polari, reflecti path

$$\vec{E} = \left(2\hat{x} + 1.5\, e^{-j\frac{\pi}{4}}\hat{y}\right) e^{-jk_0 z}\, e^{j\omega t}.$$

$$\left|\frac{E_y}{E_x}\right| = \frac{1.5}{2} = \frac{3}{4} \qquad \text{phase dif} = \frac{-\pi}{4}$$

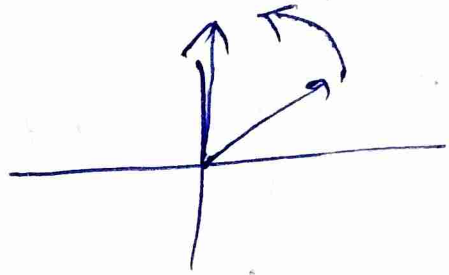
~~$\tan\phi =$~~

$$\vec{E} = 2\cos \omega t\, \hat{x} + 1.5\cos(\omega t - \frac{\pi}{4})\hat{y}.$$

$\frac{\pi}{2}$

$$\vec{E}_0 = 2\hat{x} + \frac{1.5}{\sqrt{2}}\hat{y}.$$

$$E_{\pi/4} = \frac{1.5}{\sqrt{2}}\hat{y}.$$



Left- ellipticcally polarized.

5) $\vec{J}(r') = I_0\, e^{-|z'|/a}\, \delta(x')\,\delta(y')\,\hat{z}$

5) $G(r, r') = \dfrac{e^{-jk|r-r'|}}{4\pi|r-r'|}$

$$\vec{A}(r) = \mu \int_{V'} \vec{J}(r')\, G(r, r')\, dr'$$

$$\vec{A}(r) = \mu \int_{V'} I_0\, e^{-\frac{|z'|}{a}}\, \delta(x')\,\delta(y')\, \frac{e^{-jk|r-r'|}}{4\pi|r-r'|}\, dx'dy'dz'$$

$$\Rightarrow \mu \int_{-a/2}^{a/2} I_0\, e^{-\frac{|z'|}{a}}\, \frac{e^{-jk|r-r'|}}{4\pi|r-r'|}\, dz'$$

5) $$\Rightarrow \frac{\mu I_0\, e^{-jk|r-r'|}}{4\pi|r-r'|} \int_{-a/2}^{a/2} e^{-\frac{|z'|}{a}}\, dz'$$

$$\Rightarrow \frac{2\mu I_0\, e^{-jk|r-r'|}}{4\pi|r-r'|} \int_{0}^{a/2} e^{-\frac{z'}{a}}\, dz'$$

$$\Rightarrow \frac{2\mu I_0\, e^{-jk|r-r'|}}{4\pi|r-r'|} \left(\left.\frac{e^{-\frac{z'}{a}}}{-\frac{1}{a}} \right|_{0}^{a/2} \right) \rightarrow -a\left(e^{-\frac{1}{2}} - 1\right)$$

$\Rightarrow$

$$\vec{A}(r) \Rightarrow \frac{a \mu I_0 e^{-jk|r-r'|}}{2\pi|r-r'|}\left(1-e^{-\frac{1}{2}}\right)\hat{z}.$$

c). $\nabla \cdot \vec{J} \Rightarrow$ ~~$I_0 e^{-|z'|/a}$~~

$$\Rightarrow \frac{\pm 1}{a} I_0 e^{-|z'|/a} \delta(x')\delta(y')\hat{z}$$

$$\Rightarrow -\frac{I_0}{a} \text{sgn}(z') e^{-|z'|/a} \delta(x')\delta(y')\hat{z}$$

$$\neq 0$$

$$\varphi = \frac{I_0}{aj\omega} \text{sgn}(z') e^{-|z'|/a} \delta(x')\delta(y')$$

φ varies with z' & decreases away from centre.

this will imply that $\phi \neq 0$.

d) $\vec{A}$ direction comes from the $\vec{J}$ direction.

$$\vec{A} = \int J(\vec{r}')\, g(r,r')\, dr'$$

6. $g(x, x') = \frac{-j}{2k_0} e^{-jk_0(x-x')}$

$\hookrightarrow \frac{d^2g}{dx^2} + k_0^2 g = -\delta(x-x')$

$x > x'$

a) $\frac{dg}{dx} = \frac{-j}{2k_0}(-jk_0) e^{-jk_0(x-x')} \Rightarrow \frac{-1}{2} e^{-jk_0(x-x')}$

$\frac{d^2g}{dx^2} = \frac{-1}{2}(-jk_0) e^{-jk_0(x-x')} \Rightarrow \frac{jk_0}{2} e^{-jk_0(x-x')}$

$\frac{d^2g}{dx^2} \Rightarrow -k_0^2 g$

$\therefore \frac{d^2g}{dx^2} + k_0^2 g = 0 \quad x > x'$

b) $x < x'$ $\quad g(x, x') = \frac{-j}{2k_0} e^{jk_0(x-x')}$

$\frac{dg}{dx} = \frac{-j}{2k_0}(jk_0) e^{jk_0(x-x')} \Rightarrow \frac{1}{2} e^{jk_0(x-x')}$

$\frac{d^2g}{dx^2} = \frac{jk_0}{2} e^{jk_0(x-x')} \Rightarrow -k_0^2 g$

$$\therefore \frac{d^2g}{dx^2} + k_0^2 g = 0 \quad x < x'.$$

c) $\left.\frac{dg}{dx}\right|_{x'+} - \left.\frac{dg}{dx}\right|_{x'-}$

$$\Rightarrow \left.\frac{-1}{2} e^{-jk_0(x-x')}\right|_{x=x'+} - \left.\left(\frac{1}{2} e^{+jk_0(x-x')}\right)\right|_{x=x'-}$$

$$\Rightarrow \frac{-1}{2} - \frac{1}{2} \Rightarrow -1.$$

d) $$\frac{d^2g}{dx^2} + k_0^2 g = -\delta(x-x')$$

$$\int_{x'+\epsilon}^{x'+\epsilon} \frac{d^2g}{dx^2}\,dx + \int_{x'-\epsilon}^{x'+\epsilon} \cancel{k_0^2 g\,dx}^{\,0} = -\int_{x'-\epsilon}^{x'+\epsilon} \delta(x-x')\,dx.$$

$$\int_{x'-\epsilon}^{x'+\epsilon} \frac{jk_0}{2} e^{-jk_0(x-x')}\,dx = -1$$

$$\left.\frac{dg}{dx}\right|_{x'-\epsilon}^{x'+\epsilon\ (\text{or})} = -1.$$

expanding the integral.

$$\frac{jk_0}{2} \left. \frac{e^{-jk_0|x-x'|}}{-jk_0} \right|_{x'-\varepsilon}^{x'+\varepsilon}$$

$$\Rightarrow \frac{1}{2}\left(\frac{e^{-jk_0(-\varepsilon)}}{-1} - \frac{e^{jk_0(-\varepsilon)}}{1} \right) \Rightarrow \frac{1}{2}(-1-1) \Rightarrow -1.$$

8. $g(r) = a H_0^{(1)}(kr) + b H_0^{(2)}(kr)$.

Sommerfeld radi. B.C $\Rightarrow \lim_{r\to\infty} \sqrt{r}\left(\frac{\partial g}{\partial r} + jkg\right) = 0$.

Asym. forms: $H_0^{1}(kr) \sim \sqrt{\frac{2}{\pi kr}}\, e^{j(kr - \frac{\pi}{4})}$

$$H_0^{(2)}(kr) \sim \sqrt{\frac{2}{\pi kr}}\, e^{-j(kr - \frac{\pi}{4})}.$$

a) $\frac{\partial}{\partial r} H_0^{(2)}(kr) = \sqrt{\frac{2}{\pi kr}}(-jk)\, e^{-j(kr-\frac{\pi}{4})} + \sqrt{\frac{2}{\pi kr}} \cdot \frac{1}{r} \cdot e^{-j(kr-\frac{\pi}{4})}$

$$\lim_{r\to\infty} \sqrt{r}\left(\left(\sqrt{\frac{2}{\pi kr}}\, e^{-j(kr-\frac{\pi}{4})}\left(-jk + \frac{1}{r}\right)\right) + jk\sqrt{\frac{2}{\pi kr}}\, e^{-j(kr-\frac{\pi}{4})}\right)$$

$$\lim_{r \to \infty} \sqrt{r} \sqrt{\frac{2}{\pi k r}} e^{-j(kr - \frac{\pi}{4})} \left(-jk + \frac{1}{r} + jk \right)$$

$$\Rightarrow \lim_{r \to \infty} \frac{e^{-j(kr - \frac{\pi}{4})}}{r} \sqrt{\frac{2}{\pi k}}$$

$$\Rightarrow 0.$$

5). $H_0^{(1)}(kr) \approx \sqrt{\frac{2}{\pi k r}} e^{j(kr - \frac{\pi}{4})}$

$$\frac{\partial H_0^{(1)}(kr)}{\partial r} = \sqrt{\frac{2}{\pi k r}}\, jk\, e^{j(kr - \frac{\pi}{4})} + \cancel{r}\sqrt{\frac{2}{\pi k r}}\, e^{j(kr - \frac{\pi}{4})}$$

$$\lim_{r \to \infty} \sqrt{r} \sqrt{\frac{2}{\pi k r}} e^{jk(r - \frac{\pi}{4})} \left(jk + \frac{1}{r} + jk \right)$$

$$\Rightarrow \lim_{r \to \infty} e^{j(kr - \frac{\pi}{4})} \left(2jk + \frac{1}{r} \right) \sqrt{\frac{2}{\pi k}} \Rightarrow\!\!\!\!/\; 0.$$

c) $a_k(v) = \dfrac{(\frac{1}{2}-v)_k (\frac{1}{2}+v)_k}{(-2)^k k!}$ $z = kr$

$a_0(v) = 1$. $w = z - \frac{1}{2}v\pi - \frac{\pi}{4}$.

$$H_v^{(2)}(z) \sim \left(\frac{2}{\pi z}\right)^{\frac{1}{2}} e^{-iw} \sum_{k=0}^{\infty} (-1)^k \frac{a_k(v)}{z^k}$$

$v = 0$. $H_0^{(2)}(z) \sim \left(\frac{2}{\pi z}\right)^{1/2} e^{-iw} \sum_{k=0}^{\infty} (-1)^k \frac{a_k(0)}{z^k}$.

$$a_k(0) = \frac{(\frac{1}{2}-0)(\frac{1}{2}+0)}{(-2)^k k!} \Rightarrow \frac{1}{4(-2)^k k!}$$

$a_0(0) = 1$, $a_1(0) = \dfrac{1}{-8}$ $a_2(0) = \dfrac{1}{4(4)(2)}$

$$H_0^{(2)}(z) = \left(\frac{2}{\pi z}\right)^{1/2} e^{-j(kr - \frac{\pi}{4})} \left(\frac{1}{1} + \frac{(-1)}{kr}\left(\frac{-1}{8}\right)\right)$$

$$H_0^{(2)}(z) = \left(\frac{2}{\pi z}\right)^{1/2} e^{-j(kr - \frac{\pi}{4})} \left(1 + \frac{1}{8kr}\right)$$

↳ two-term expansion.

$\frac{\partial H_0^{(2)}(kr)}{\partial r}$ of second term

$$\Rightarrow \sqrt{\frac{2}{\pi k r}}\, e^{-j(kr-\frac{\pi}{4})} \left(\frac{1}{8kr}\right)$$

$\frac{\partial H_0^{(2)}(kr)}{\partial r}$ of 2nd term

$$\Rightarrow \sqrt{\frac{2}{\pi k r}} \cdot \frac{1}{k \cdot 8 r^2}\, e^{-j(kr-\frac{\pi}{4})} + \sqrt{\frac{2}{\pi k r}} \cdot \frac{(-jk)}{8kr}\, e^{-j(kr-\frac{\pi}{4})}$$

$$\Rightarrow \frac{1}{8kr}\sqrt{\frac{2}{\pi k r}}\, e^{-j(kr-\frac{\pi}{4})}\left(\frac{1}{r} - jk\right).$$

const $= \alpha$.

$$\lim_{r\to\infty} \frac{\sqrt{r}}{r\sqrt{r}}(\alpha)\, e^{-j(kr-\frac{\pi}{4})}\left(\frac{1}{r} - jk + jk\right).$$

$$\lim_{r\to\infty} \alpha \frac{e^{-j(kr-\frac{\pi}{4})}}{r^2} = 0.$$

2nd term $\to 0$, 1rst term $\to 0$ from (a).

d) Hankel functions are just complex ~~[illegible]~~ variation of Bessel function, ~~[illegible]~~ used to better represent the standing waves.

9. 1D: $g_{1D} = \frac{-j}{2k} e^{-jk|x-x'|}$

Far field

2D: $g_{2D} = \frac{-j}{4} H_0^{(2)}(k|\vec{r}-\vec{r}'|)$, $|g_{2D}| \sim \frac{1}{\sqrt{2\pi k r}}$ $r \gg \lambda$

3D: $g_{3D} = \frac{e^{-jk|r-r'|}}{4\pi|r-r'|}$

(a) $|g_{1D}| = \frac{1}{2k}$ ← Ampl. decay fn vs. dist.

$\Rightarrow \frac{1}{2k} \rightarrow$ const.

$|g_{2D}| \Rightarrow \sqrt{\frac{2}{\pi k 4|r-r'|}} \leftrightarrow \frac{1}{\sqrt{r}}$ dependency

$|g_{3D}| = \frac{1}{4\pi|r-r'|} \rightarrow \frac{1}{r}$ dependancy

total. power (cons).

1D $\rightarrow$ plane wave $\rightarrow$ energy const.

$I = \frac{P}{Area}$ $\therefore$ amplitude = const.

2D $\rightarrow$ spreads on circle. $\rightarrow$ circumference $\rightarrow$ ~~energy spreads~~ to $2\pi r$

$\therefore$ amplitude $\propto \frac{1}{\sqrt{r}}$

3D $\rightarrow$ spreads as spherical $\rightarrow$ area ~~energy~~ $\Rightarrow 4\pi r^2$

amplitude $\propto \frac{1}{r}$.

Decay law $\Rightarrow \frac{1}{r^{D-1}}$ D = dimension

b). Energy is conserved.

Intensity $(I) = \frac{P}{Area}$.

$I \propto$ Amplitude^2 (A^2)

$A \propto \sqrt{I} \propto \frac{1}{\sqrt{Area}}$.

1D. $\rightarrow$ $A \propto$ constant
(const Area)

2D $\rightarrow$ $A \propto \frac{1}{\sqrt{2\pi r}}$ $\Rightarrow A \propto \frac{1}{\sqrt{r}}$.

3D $\rightarrow$ $A \propto \frac{1}{\sqrt{4\pi r^2}}$ $\Rightarrow A \propto \frac{1}{r}$.

c). Near field is given as $kr << 1$ and far field $kr >> 1$, with the experimental or statistical estimation the transition distance is $\frac{2D^2}{\lambda}$.
3D Green's function describes it

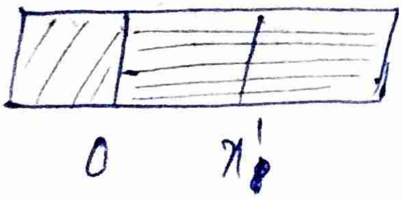
d) 1D Green's function $\rightarrow$ no amplitude decay.
There is no propagation / spread in other direction, its similar to the plane wave propagation in one direction without attenuation

Real antennas show $1/r$ decay. bz in the far field they spread like a sphere with $4\pi r^2$ area and hence

$$A \propto \frac{1}{r} \text{ decay.}$$

10. $$\frac{d^2 g}{dx^2} + k^2(x) g = -\delta(x - x'),$$

$$k(x) = \begin{cases} k_1 = \omega\sqrt{\mu_0 \varepsilon_0 \varepsilon_{r1}} & x < 0 \\ k_2 = \omega\sqrt{\mu_0 \varepsilon_0 \varepsilon_{r2}} & x > 0 \end{cases}$$



$x \to \pm\infty$ ~~waves should die out~~

a) $x < 0$ $\quad \frac{d^2 g}{dx^2} + k^2(x) g = 0.$

$$g = Ae^{-jk_1 x} + Be^{+jk_1 x} \qquad x \to -\infty$$

$g = Be^{+jk_1 x}.$ $\quad \therefore A = 0.$ wave should travel left

$0 < x < x' \Rightarrow Ce^{-jk_2 x} + De^{jk_2 x}$

$x > x' \Rightarrow Fe^{+jk_2 x} + Ee^{-jk_2 x}.$ $\quad x \to \infty$

$\Rightarrow Ee^{-jk_2 x}.$ Right travelling wave

b) g , $\frac{dg}{dx}$ are contin. at $x = 0$.

g contin. at $x = x'$.

$g_I(0) = g_{II}(0)$ $\quad\quad$ $Bjk_1 = -j(k_2 + jDk_2$.

$B = C + D$. —(1) $\quad\quad$ $\frac{Bk_1}{k_2} = (-C + B)$ —(2)

$$Ce^{-jk_2x'} + De^{jk_2x'} = Ee^{-jk_2x'}$$

$$C + De^{2jk_2x'} = E. \text{ —(3)}$$

$$\int_{x'-}^{x'+} \frac{d^2g}{dx^2}\,dx + \int_{x'-}^{x'+} \cancel{k^2 g\,dx}^{\,\to 0} = -\int_{x'-}^{x'+} \delta(x-x')\,dx.$$

$$\left.\frac{dg}{dx}\right|_{x'-}^{x'+} = -1.$$

$$-jk_2Ee^{-jk_2x'} - \left(-jCk_2e^{-jk_2x'} + jk_2De^{jk_2x'}\right) = -1$$

$$C - De^{2jk_2x'} = \frac{je^{jk_2x'}}{k_2} + E. \text{ —(4)}$$

③ − ④ $\quad 2D e^{2jk_2x'} = -\dfrac{j e^{jk_2x'}}{k_2}$

$$D = -\frac{j e^{-jk_2x'}}{2k_2}$$

③ + ④ $\quad C = \dfrac{j e^{jk_2x'}}{2k_2} + E$

① + ② $\quad B\left(\dfrac{k_1+k_2}{k_2}\right) = 2D$

$$B = \frac{2k_2}{k_1+k_2}\left(\frac{-j e^{-jk_2x'}}{2k_2}\right) \Rightarrow \frac{-j e^{-jk_2x'}}{k_1+k_2}$$

① − ② $\quad B\left(\dfrac{k_2-k_1}{k_2}\right) = 2C$

$$C = \left(\frac{k_2-k_1}{2k_2}\right)\left(\frac{-j e^{-jk_2x'}}{k_1+k_2}\right) \Rightarrow \Gamma D$$

$$E \Rightarrow \Gamma\left(\frac{-j}{2k_2}\right) e^{-jk_2x'} - \frac{j}{2k_2} e^{+jk_2x'}$$

$$g \Rightarrow \frac{-j}{2k_2}\left(e^{-jk_2(x-x')} + \Gamma e^{-jk_2(x+x')}\right) \quad x>x'$$

$$g \Rightarrow \frac{-j}{2k_2}\left(e^{-jk_2|x-x'|} + \Gamma e^{-jk_2(x+x')}\right) \quad x'>x>0$$

$$g \Rightarrow \frac{-j}{k_1+k_2} e^{-jk_2x'} e^{jk_1x} \quad x<0.$$

$\Rightarrow$

c) $\Gamma = \frac{k_2-k_1}{k_1+k_2}$ $\quad \varepsilon_{r1} = \varepsilon_{r2}$

$k_1 = k_2 = k$

$\Gamma = 0.$

$$g = \frac{-j}{k} e^{-jk|x-x'|}.$$

d) $\varepsilon_{r1} = 1, \quad \varepsilon_{r2} = 4.$ $\quad k \propto \sqrt{\varepsilon_r}$

$\cancel{\Phi = \frac{4-1}{5} = \frac{3}{5}}$ $\quad k_1 = k_0 \quad k_2 = 2k_0.$

$$\Gamma = \frac{2k_0 - k_0}{3k_0} = \frac{1}{3}.$$

$|g|$ inside medium 2 produce standing wave and oscillations because of both transmission & reflection, whereas $|g|_{homogen}$ is always const

Amplitude transmitted $=$ $+$ $(1+\Gamma)$
$x = 0$

$\Rightarrow \frac{1}{3} + 1 \Rightarrow \frac{4}{3}.$